Integrated Paleomagnetism and U-Pb Geochronology of Mafic Dikes of the Eastern Anabar Shield Region, Siberia: Implications for Mesoproterozoic Paleolatitude of Siberia and Comparison with Laurentia

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ABSTRACT

This article reports the first joint paleomagnetic and U-Pb geochronologic study of Precambrian diabase dikes in the Anabar Shield and adjacent Riphean cover of Siberia. It was undertaken to allow comparison with similar published studies in Laurentia and to test Proterozoic reconstructions of Siberia and Laurentia. An east-trending Kuonamka dike yielded a provisional U-Pb baddeleyite emplacement age of 1503 ± 5 Ma and a virtual geomagnetic pole at 16°S, 221°E (dm = 17°, dp = 10°). A paleomagnetic pole at 6°N, 234°E (dm = 28°, dp = 14°) was obtained from five Kuonamka dikes. An east-southeast-trending Chieress dike yielded a U-Pb baddeleyite emplacement age of 1384 ± 2 Ma and a virtual geomagnetic pole at 4°N, 258°E (dm = 9°, dp = 5°). Kuonamka and Chieress poles are interpreted to be primary but do not average out secular variation. Assuming that the Siberian Plate has remained intact since the Mesoproterozoic, except for mid-Paleozoic opening of the Viljuy Rift, then the above results indicate that the Siberian Plate was in low latitudes at ca. 1503 and 1384 Ma, broadly similar to low latitudes determined for Laurentia from well-dated paleopoles at 1460–1420, 1320–1290, and 1267 Ma. This would allow Laurentia and Siberia to have been attached in the Mesoproterozoic, as suggested in several recent studies based on geological criteria. However, because paleomagnetic results from the Anabar Shield region do not average out secular variation and the ages of poles from Siberia and Laurentia are not well matched, it is not yet possible to distinguish between these reconstructions or to rule out other configurations that also maintain the two cratons at low paleolatitudes.

Introduction

Mesoproterozoic to Neoproterozoic continental reconstructions involving Siberia are a topic of much debate, often in the context of the postulated supercontinent of Rodinia. An early model based on geological correlations placed the Siberian Plate adjacent to southwestern Laurentia (present coordinates) in the Mesoproterozoic (fig. 1*a*; Sears and Price 1978). More recently, a number of Meso-Neoproterozoic reconstructions have located Siberia next to northern Laurentia (e.g., fig. 1*b*–1*e*; Hoffman 1991; Condie and Rosen 1994; Pelechaty 1996; Frost et al. 1998; Rainbird et al. 1998). Some are based on the comparison of Paleoproterozoic or even Archean geological features between the two continental blocks. However, the relative orientation of the two blocks in figure 1*b*–1*e* shows dramatic differences, and these geology-based reconstructions remain highly controversial (e.g., Khudoley 1997; Rainbird and de Freitas 1997; Frost et al. 1998; Rainbird et al. 1998).

Attempts to use paleomagnetism to determine the relative location of Laurentia and Siberia in the late Precambrian have also yielded dramatically divergent results. For example, Poorter (1981) placed Siberia adjacent to southern Baltica and the British Isles in the "Late Precambrian/Cambrian" (fig. 1*f*). Piper (1982) located Siberia adjacent to southwestern Laurentia (similar to the reconstruction by Sears and Price [1978] in fig. 1*a*), using pre-850-Ma Siberian paleomagnetic data. Scotese and McKerrow (1990) proposed that Siberia lay near eastern Greenland in the latest Precambrian (fig. 1*g*).

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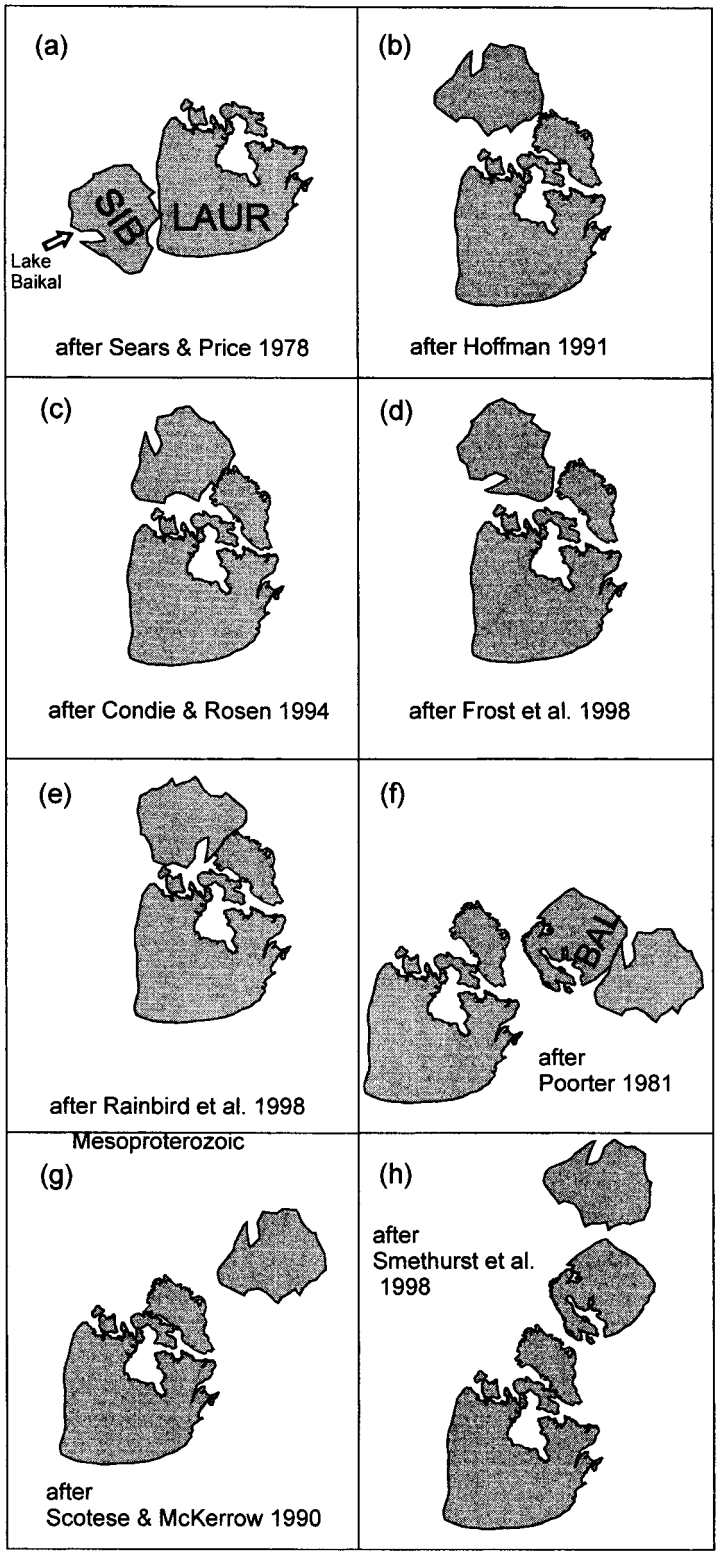


Figure 1. Examples of Precambrian reconstructions involving Siberia, Laurentia, and Baltica from the literature. The corresponding labels *SIB*, *LAUR*, and *BAL* are aligned with respect to present-day east-west. The times for which these reconstructions apply are discussed in the text. For ease of comparison, Laurentia is shown in a consistent orientation throughout. Siberia and Greenland are located as closely as possible to their position in the original reference. However, in some cases, there is minor distortion because of the projection in the original.

Recently, Smethurst et al. (1998) used paleomagnetic data to place Siberia adjacent to eastern Baltica rather than Laurentia (fig. 1*h*) or, alternatively, near East Antarctica in a 750-Ma Rodinia reconstruction. Unfortunately, Precambrian paleopoles from Siberia are usually so poorly dated, often with uncertainties of hundreds of millions of years, that paleomagnetic reconstructions, like those based on geology, are suspect.

Recent integrated studies of paleomagnetism and U-Pb baddeleyite and zircon geochronology on mafic dikes and sills (e.g., Buchan et al. 1998) have demonstrated that it is possible to obtain Precambrian paleopoles with high-precision ages. In order to better constrain the paleolocation of Siberia for use in paleocontinental reconstructions and to test published reconstructions involving Siberia, we have studied the paleomagnetism and U-Pb geochronology of diabase dikes from a river traverse across the eastern Anabar Shield and adjacent Riphean covered craton of Siberia.

Geological Setting

The Siberian Plate (fig. 2) is underlain mainly by Archean crust, except in the Paleoproterozoic Akitkan fold belt that may mark the site of Paleoproterozoic convergence (Rosen et al. 1994) and in Paleoproterozoic portions of the Aldan Shield (Frost et al. 1998). Timing of amalgamation of Archean blocks of the Siberian Plate is controversial. Khramov et al. (1985), on the basis of a comparison of paleomagnetic apparent polar wander paths, proposed that the Siberian Plate consisted of at least three blocks—the northern, southwestern, and southeastern portions of the plate—which moved relative to one another until at least the Cambrian. Pavlov and Petrov (1997), using paleomagnetism, argued for the integrity of the Siberian Plate since at least the Middle Riphean (ca. 1350–1000 Ma). Zonenshain et al. (1990) suggested that at least five continental blocks—"Angara-Anabar" (*A* and *M* in fig. 2), Tungus, Olenek, "Aldan" (*OL*, *AL*, *U*, and *B* in fig. 2), and Stanovoy—were separated by ocean basins of unknown widths in the Paleoproterozoic but had amalgamated to form a single Siberian Plate by Early Riphean (ca. 1650–1350 Ma) time. Rosen et al. (1994) summarized geochronological evidence for widespread 2000–1800-Ma orogenic events in the Siberian Plate. They concluded that the major event or events at 1950–1900 Ma, based on ages of major fault zones, correspond to its time of assembly. Hoffman (1991) proposed that the Siberian Plate lay adjacent to present-day northern Laurentia at that time because the events in Siberia are coeval with similar events that mark the assembly of northern Laurentia.

Most of the Siberian Plate is covered by Riphean (ca. 1650–650 Ma), Vendian, and Phanerozoic sediments or by the 250-Ma Siberian Traps flood basalt province. However, Archean basement is exposed in two large Precambrian shield areas, the Aldan and Anabar shields, as well as several smaller windows (fig. 2; e.g., Condie and Rosen 1994; Rosen et al. 1994).

Aulacogens of at least two ages, Riphean and Devonian, are present in the Siberian Plate (Zonenshain et al. 1990). The younger Viljuy aulacogen (fig. 2), consisting of the Viljuy (G1) and Ygyatinsky (G2) grabens, extends westward ~600 km into the Siberian Plate (Zonenshain et al. 1990). Analysis of paleomagnetic data (Gurevich 1984; Pavlov and Petrov 1996, 1997; Smethurst et al. 1998) suggests a counterclockwise rotation of the portion of the Siberian Plate north of the Viljuy aulacogen relative to that south of the aulacogen in the mid-Paleozoic (Wenlock-Esmian: 0.43–0.39 Ga). Smethurst et al. (1998) estimated a rotation of ~20° about a rotation pole at 60°N, 100°E, just west of the basin.

Proterozoic diabase dikes are abundant in the main areas of exposed basement in the Siberian Plate and in Riphean and Vendian cover rocks (Ernst et al. 1996 and references therein). In an earlier study, those in and adjacent to the Anabar Shield were divided into 10 swarms on the basis of trend, location, composition, and K-Ar age (fig. 3; Okrugin et al. 1990). Five of these swarms occur in the general area of this study. The east-trending Kuonamka swarm (K-Ar ages of 1600–1200 Ma), the southeast-trending East Anabar swarm (K-Ar ages of ca. 1700 Ma), and the east-northeast-trending Central Anabar swarm (K-Ar ages of 1800–1400 Ma) intrude highly metamorphosed Archean rocks and, occasionally, Early Riphean rocks. The Chieress swarm (K-Ar ages of ca. 1200–900 Ma) and the Juken swarm (K-Ar ages of 1300–1200 Ma), both of east-southeast trend, cut Early Riphean cover rocks immediately to the east of the Anabar Shield.

Sampling and Laboratory Procedure

Eleven diabase dikes were sampled for paleomagnetism, and of these, eight were also sampled for U-Pb geochronology, during the course of a rafting expedition down the Bol'shaya Kuonamka River in the eastern Anabar Shield region (fig. 3). Each dike is assigned to one of the swarms described above, based on an analysis of its trend, petrographic and/or geochemical characteristics, paleomagnetic direction, and U-Pb or previously published K-Ar age.

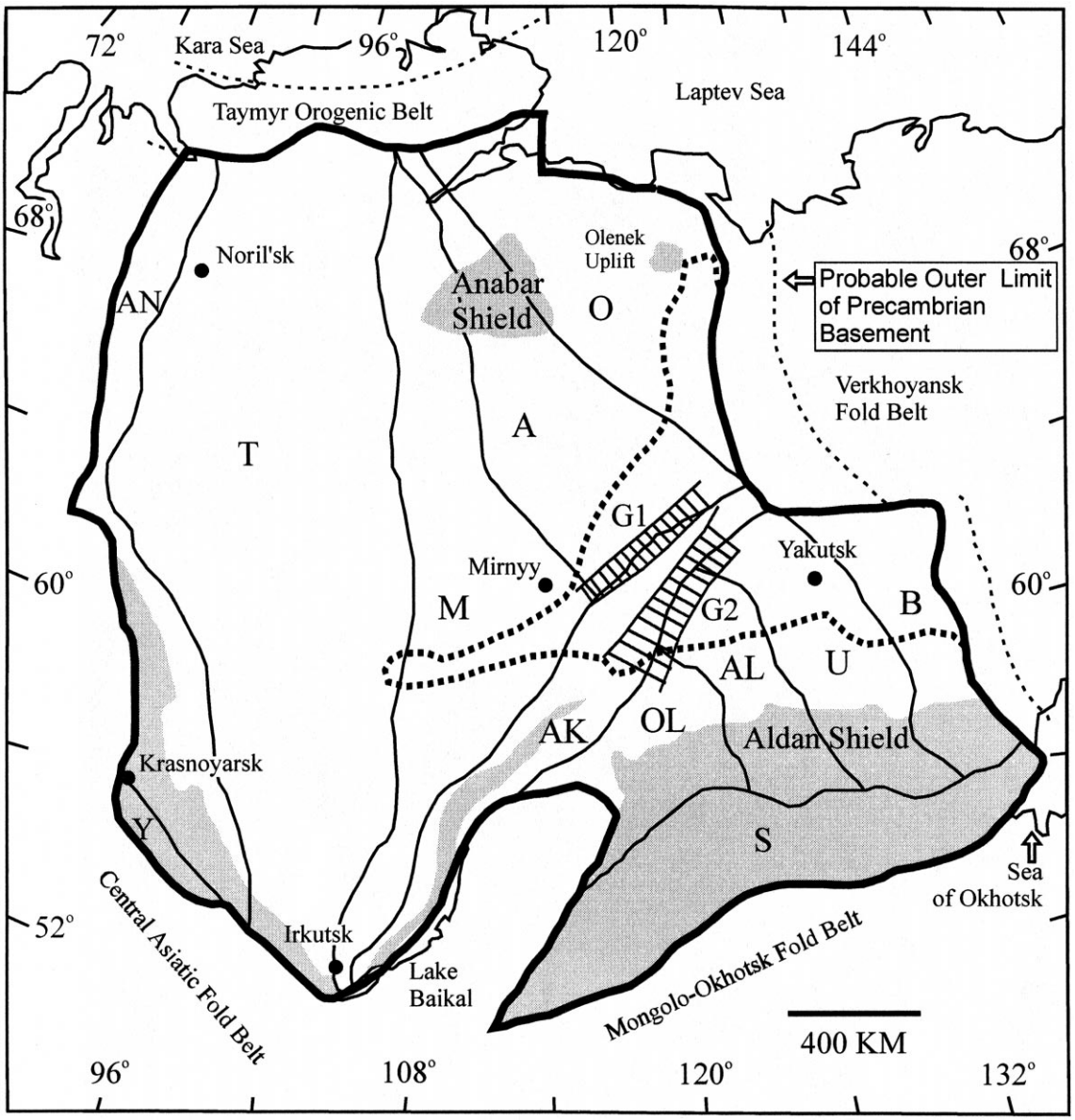


Figure 2. Precambrian crustal provinces of the Siberian Plate (after Rosen et al. 1994). Belts with Archean basement (after Rosen et al. 1994) are Aldan (*AL*), Anabar (*A*), Angara (*AN*), Bayomga (*B*), Magan (*M*), Olekma (*OL*), Olenek (*O*), Stanovoy (*S*), Tungus (*T*), Uchur (*U*), and Yenisey (*Y*). Akitkan (*AK*) is a Paleoproterozoic fold belt. *G1* and *G2* are the Ygyatinsky and Viljuy grabens, respectively, containing 10–12 km of Upper Devonian to Lower Carboniferous sediments and 380–350-Ma volcanics and capped by Mesozoic-Cenozoic sediments of the Viljuy Basin (*heavily dotted outline*) (Zonenshain et al. 1990). Shading denotes areas containing extensive exposures of Precambrian shield and Riphean cover.

Thus, it is thought that dikes at sites 1, 2, 3, 4, and 14 belong to the Kuonamka swarm; at site 10 to the Central Anabar swarm; at site 11 to the East Anabar swarm; at sites 13, 15, and 16 to the Juken swarm; and at site 17 to the Chieress swarm.

At most of the paleomagnetic sites, six or seven block samples were oriented with both solar and magnetic compasses. However, only three samples could be obtained at site 1, of which two were oriented by magnetic compass. In addition, topographic sightings were used instead of the sun at sites 13, 14, and 16. In the laboratory, each oriented sample was drilled to produce two or more cylindrical specimens with a diameter of 2.5 cm and a

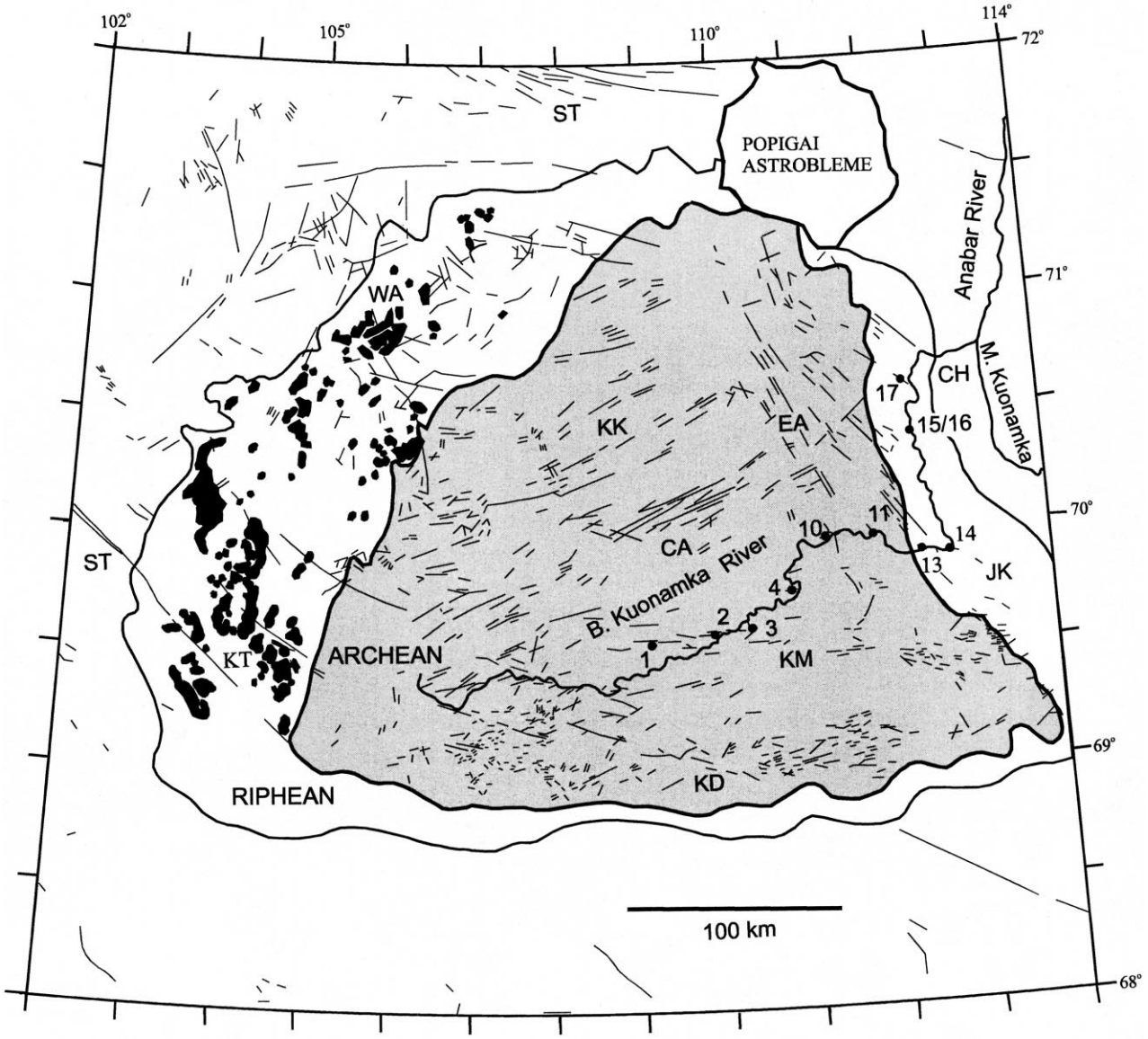


Figure 3. Diabase dikes and sills of the Anabar Precambrian Shield and vicinity (after Okrugin et al. 1990; A. V. Okrugin, unpub. data) and sampling sites. Dike swarms are *CA* = Central Anabar, *CH* = Chieress, *EA* = East Anabar, *JK* = Juken, *KD* = Kengede, *KK* = Kotuykan, *KM* = Kuonamka, *KT* = Kotuy, *ST* = Siberian Traps, and *WA* = West Anabar; 1300–900-Ma sills are shown in black. The 36-Ma (Bottomley et al. 1997) Popigai astrobleme is located on the northeast margin of the Shield. *B. Kuonamka* and *M. Kuonamka* are the Bol'shaya and Malaya Kuonamka Rivers, respectively.

length of ca. 2.3 cm. Remanence measurements were carried out using a Schonstedt DSM-1 spinner magnetometer. In general, alternating field (AF) demagnetization of individual specimens was performed in stepwise fashion to peak fields of 2.5, 5.0, 7.5, 10.0, 12.5, 15.0, 17.5, 20.0, 25.0, 30.0, 35.0, 40.0, 50.0, 60.0, 70.0, 80.0, 90.0, and 100.0 mT using a Schonstedt GSD-1 demagnetizer. Thermal demagnetization experiments were usually carried out at peak temperatures of 200°, 300°, 400°, 425°, 450°, 475°, 500°, 520°, 540°, 550°, 557°, 564°, 571°, 578°, 585°, and 592°C using a Schonstedt thermal demagnetizer, modified to maintain peak temperature control of ±5°. A few samples were also demagnetized at 610°, 630°, and 650°C. In some cases, stepwise AF demagnetization was carried out before the thermal cleaning experiments in order to remove large, soft components of probable viscous origin. All of the above experiments were conducted in a magnetically shielded room with resid-

ual magnetic field of <3000 nT. Paleomagnetic data were analyzed using the Kirschvink (1980) method and the remagnetization circle method (Bailey and Halls 1984). It was assumed that post-dike-emplacement crustal tilting is minimal in the sampling area, so that no structural correction was necessary to the paleomagnetic data. This was confirmed at sites 2, 4, and 11 along the upper portion of the river traverse, despite poor outcrop, by magnetic fabric studies (R. E. Ernst, unpub. data), which can test for vertical dike dip (Ernst and Baragar 1992), and, in the lower portion of the river traverse, by subhorizontal host Riphean sedimentary rocks.

Three samples, one each from sites 2 and 4 of the Kuonamka swarm and one from Chieress site 17, were selected for U-Pb analysis. Approximately 6–10 kg of sample from each site were processed for mineral separation and hand picking. Heavy minerals were concentrated from representative samples via conventional crushing, grinding, Wilfley table, and heavy-liquid techniques. Purification of the heavy mineral concentrates was carried out using a Frantz LB-1 isodynamic separator. Baddeleyite grains were sought and found in the least-magnetic fractions of two of the three samples (sites 4 and 17). Best-quality-selected baddeleyites were given brief ultrasonic treatment in ethyl alcohol in order to remove minor adhering quartz and feldspar fragments before dissolution. Highest-quality zircon (from site 2) was handpicked on the basis of morphology, optical clarity, rarity of cracks, and absence of inclusions or apparent cores. In all cases, zircons were given strong air abrasion before analysis. The U-Pb analytical techniques used in this study are broadly similar to those outlined by Parrish et al. (1987), while treatment of analytical errors follows Roddick (1987), with regression procedures modified after York (1969). Analytical results, keyed to lettered fractions in the concordia diagrams, are presented below. Errors on the ages are presented in the text and in concordia plots at 95% confidence limits (2σ).

Petrography

East-trending Kuonamka dikes at sites 1, 2, 3, 4, and 14 are diabases consisting dominantly of plagioclase and clinopyroxene, with minor Fe-Ti oxides. In fine-grained portions the fabric is intergranular, and in medium-grained portions the fabric is subophitic. The mineralogy at these sites is fresh except for pervasive sausseritization of plagioclase at site 14, minor alteration of clinopyroxene at site 3, and deuteric replacement of a trace second ferromagnesian mineral at site 1. Myrmekite is observed at site 4 and trace quartz is observed at site 14. Site 3 contains fine needles of apatite, and site 14 is cut by a few small aplite (rheomorphic) dikes.

The 75-m-wide east-southeast-trending Chieress dike at site 17 consists of a marginal phase of gabbro and a 50-m-wide interior pulse of monzodiorite and is cut by a late-stage aplite dike. All paleomagnetic samples were from the monzodiorite phase since no material of the other phases was found in situ. The gabbroic phase consists of partially sausseritized plagioclase and weakly uralitized clinopyroxene in an intergranular texture and also opaques (Fe-Ti oxides) and minor apatite. The clinopyroxene is partially replaced by pumpellyite, actinolite, and biotite. The monzodiorite phase contains feldspar that has been pervasively sausseritized and hornblende that has been totally replaced by pumpellyite and actinolite. Minor apatite and quartz are also present. The aplite dike consists of feldspar phenocrysts in a fine-grained quartz-feldspar matrix. The presence of coexisiting pumpellyite and actinolite in both the gabbroic and monzodioritic phases at site 17 implies subgreenschist-grade metamorphism.

Geochemical studies of Juken dikes with variable trends at sites 15 and 16 and at the probable continuation of the east-southeast-trending dike from site 13 revealed high-K compositions (Okrugin et al. 1990; A. V. Okrugin, unpub. data). The dike at site 13 is an altered diabase consisting of highly sausseritized plagioclase, chlorite after clinopyroxene, Fe-Ti oxides, and minor apatite. Its continuation contains scattered amygdules filled with calcite. Sites 15 and 16 are from a dike and a pod (or neck), respectively, at a single locality and probably fed an overlying lava flow ~10 m above. Texturally, these intrusions are volcanic breccias containing amygdules (with chorite and/or calcite), xenoliths of quartz, lithic fragments, hematized volcanic glass, and a hematized and/or chloritized groundmass. Microprobe analysis (A. V. Okrugin, unpub. data) indicates that the groundmass consists of potassium feldspar grains (SiO_2 62%–65%, Al_2O_3 17.5%–18.5%, FeO 0.1%–1%, K_2O 16%–18%, and CaO and $Na_2O < 1\%$) in a chlorite matrix (SiO_2 31%–32%, Al_2O_3 12.5%–13.5%, FeO 11%–15%, MgO 21%–26%, and OH [calculated] 11%–12%).

The east-northeast-trending dike at site 10, assigned to the Central Anabar swarm, is a fresh diabase consisting dominantly of plagioclase and clinopyroxene with Fe-Ti oxides and fine needles of apatite. The diabase dike at site 11, belonging to the East Anabar swarm, contains distinctive greenish (sausseritized) plagioclase phenocrysts. The groundmass is fresh and comprises mainly plagio-

clase and clinopyroxene with minor Fe-Ti oxides. More generally, the East Anabar swarm is characterized by plagioclase phenocrysts (A. V. Okrugin, unpub. data) and minor olivine (Okrugin et al. 1990); however, olivine was not observed at site 11.

Paleomagnetic Results

At most paleomagnetic sampling sites a stable remanence was isolated on stepwise AF or thermal demagnetization after elimination of a steep down component, which is probably of viscous origin. However, in some sites, systematic directional swings at higher fields or temperatures in the demagnetization experiments reflect the presence of two stable components. Paleomagnetic directional data at each site are cataloged in table 1 and plotted in figure 4. Results from each dike swarm are described below.

Kuonamka Dikes. East-trending dikes at sites 1, 2, 3, 4, and 14 are assigned to the Kuonamka swarm (fig. 3). Their mean remanence directions are shallow and east to northeast (fig. 4). The large directional scatter may represent secular variation of a single swarm, although the presence of more than one age of dike cannot be ruled out. Sites 2 and 14 exhibit stable northeast remanences with shallow inclination after elimination of steep viscous components (fig. 5*a*, 5*b*). In each of these sites the characteristic component was readily isolated using Kirschvink (1980) analysis. A shallow south-southwesterly magnetization direction was obtained at site 3 (fig. 5*c*). The remanence at site 3 is approximately reversed to those at sites 2 and 14, although some samples from site 3 showed evidence of a normally magnetized component as well. At site 1, two of the three samples yielded shallow northeasterly magnetization directions, whereas the third sample, collected about 200 m away from the others along the same dike, was magnetized with the opposite polarity.

At site 4, remanence directions follow long swings along great circle paths on AF demagnetization (fig. 6). Remagnetization circle analysis (Bailey and Halls 1984) yielded a well-defined easterly intersection, representing the most stable component of magnetization. A single sample was thermally demagnetized and also gave a long swing to the easterly direction.

At most Kuonamka sites, blocking temperatures (T_B's) of the stable component fall within the 500°–585°C range, indicating that low-titanium titanomagnetite is the dominant carrier of remanence. For example, at site 4, the stable easterly component defined by the intersection of great circles has T_B's of 520°–550°C. At site 1, however, T_B's are significantly lower, mostly between 300° and 450°C. At site 3 the T_B range is quite broad, from 200° to >580°C (fig. 5*c*). In some specimens at this site, a semistable direction is maintained to temperatures of 630°C, indicating the presence of hematite or titanohematite (fig. 5*c*).

Chieress Dike. On stepwise thermal demagnetization, the Chieress paleomagnetic sampling site (site 17) yields a stable shallow north-northeast remanence, with most T_B's between 400° and 570°C, after the elimination of a steep down viscous component in peak temperatures of 400°C or less (fig. 7*a*). On the basis of these blocking temperatures, the dominant carrier of the stable remanence is thought to be low-titanium titanomagnetite. In one sample, a less stable but roughly similar direction is maintained between 570° and 630°C (fig. 7*a*), indicating the presence of hematite.

On AF demagnetization, the data are more complicated because coercivities of the steep down component significantly overlap those of the characteristic component. As a result, directions swing along a great circle away from the direction of the present earth's magnetic field toward the shallow up northeast direction, then back toward the present field before returning once again to the shallow up northeast direction (fig. 7*b*). In most instances the stable characteristic component direction is obtained in the higher coercivity steps.

Because of their relative simplicity, the thermal demagnetization data were used for statistical calculations of the mean direction at site 17 (table 1). Since only a single dike was sampled, secular variation has not been averaged out.

Juken Dikes. The Juken swarm was sampled for paleomagnetism at two localities. At one locality, two small irregular intrusions (sites 15 and 16) yielded very stable shallow up directions to the south-southwest (fig. 8*a*, 8*b*). Most T_B's fall between 300° and 585°C (fig. 8*b*), suggesting low-titanium titanomagnetite as the dominant carrier of remanence. However, in some samples (not shown), a low-intensity component with blocking temperatures >585°C indicates the presence of hematite as well.

At site 13 most samples have a high-coercivity/high-T_B component directed to the northeast with a shallow up inclination (fig. 8*c*) and a steep down component, likely of viscous origin. The two components are best isolated using a combination of AF and thermal demagnetization. However, in some cases the T_B's and coercivity spectra are so overlapped that separation of the two components is difficult (fig. 8*c*). Juken sites 15 and 16 have been

Table 1. General Characteristics and Paleomagnetic Results for Dikes of the Eastern Anabar Shield Region

Site	Swarm	Location	Trend	Width (m)	U-Pb age ($\pm 2\sigma$) (Ma)	K-Ar age ($\pm 2\sigma$) (Ma)	N, n	A, T	D, I (°, °)	D', I' (°, °)	α_{95} (°)	k	Virtual geomagnetic pole
1	KU	69°32.7′N, 109°07.9′E	E	0.7	…	…	3, 3	A	55.7, −8.9	57.4, −7.3	20	38	…
2	KU	69°35.8′N, 109°55.0′E	E	20	2728–2679 (xenocrysts)	1242 ± 20	6, 5	A	65.6, −8.4	66.6, −7.4	20	16	…
3	KU	69°37.0′N, 110°18.2′E	E	30	…	1487 ± 10	6, 5	A	207.1, −18.4	207.7, −19.2	7	117	…
4	KU	69°45.8′N, 110°50.6′E	E	50	1503 ± 5	1391 ± 42	6, 6	A	85.5, −34.8	85.6, −34.8	12/18	75	16°S, 211°E (dm = 17°, dp = 10°)
10	CA?	69°58.4′N, 111°15.7′E	ENE	30	…	…	6, 4	A, T	173.4, −7.9	172.6, −8.0	16	32	…
11	EA	69°59.2′N, 111°50.7′E	SE	30	…	1676 ± 32	6, 0	…	…	…	…	…	…
13	JU	69°54.4′N, 112°35.9′E	ESE	30–40	…	1200 ± 8	6, 4	A, T	52.7, −13.0	51.2, −13.7	20	22	…
14	KU	69°53.8′N, 112°48.0′E	E	35	…	1537 ± 34	7, 5	A, T	51.6, −13.6	49.9, −14.4	10	61	…
15/16	JU?	70°23.9′N, 112°22.8′E	Var.	1–5	…	924 ± 2 1359 ± 14 1416 ± 4	12, 8	A, T	220.2, −24.6	218.9, −23.5	7	61	…
17	CH	70°37.4′N, 112°20.5′E	ESE	75	1384 ± 2	1201 ± 34 984 ± 14	7, 7	T	34.3, −23.8	33.1, −25.1	9	50	4°N, 258°E (dm = 9°, dp = 5°)
Mean[a]	…	…	…	…	…	…	5	…	…	56.8, −9.5	28	9	6°N, 234°E (dm = 28°, dp = 14°)

Note. Swarm names are CA = Central Anabar, CH = Chieress, EA = East Anabar, JU = Juken, KU = Kuonamka (after Okrugin et al. 1990). Trend and width of the dike were measured at the sampling site. "Var." stands for variable trend. K-Ar ages (K) are from Okrugin et al. (1990) and A. Okrugin (unpub. data) and were determined at the geochronology laboratory of the Yakutian Institute of Geosciences. U-Pb ages (U) are from this study. N and n are the number of oriented samples that were measured and the number that were used in the site mean calculation, respectively. A or T indicates whether thermal or AF demagnetization data were used to calculate the site mean direction. D and I are declination and inclination of the mean site direction, respectively. D and I at site 4 were determined from remagnetization circle analysis (see text). D', I' are the declination and inclination recalculated for reference locality: 70°N, 111°E. α_{95} is the radius of the circle of 95% confidence about the mean site direction. k is the precision parameter. Uncertainties in paleolatitude and paleo-orientation derived from virtual geomagnetic poles that do not average out secular variation are 10° and 18° for Kuonamka site 4; 14° and 28° for the mean of Kuonamka sites 1, 2, 3, 4, and 14; and 5° and 10° for Chieress site 17.

[a] Mean of KU sites 1, 2, 3, 4, and 14.

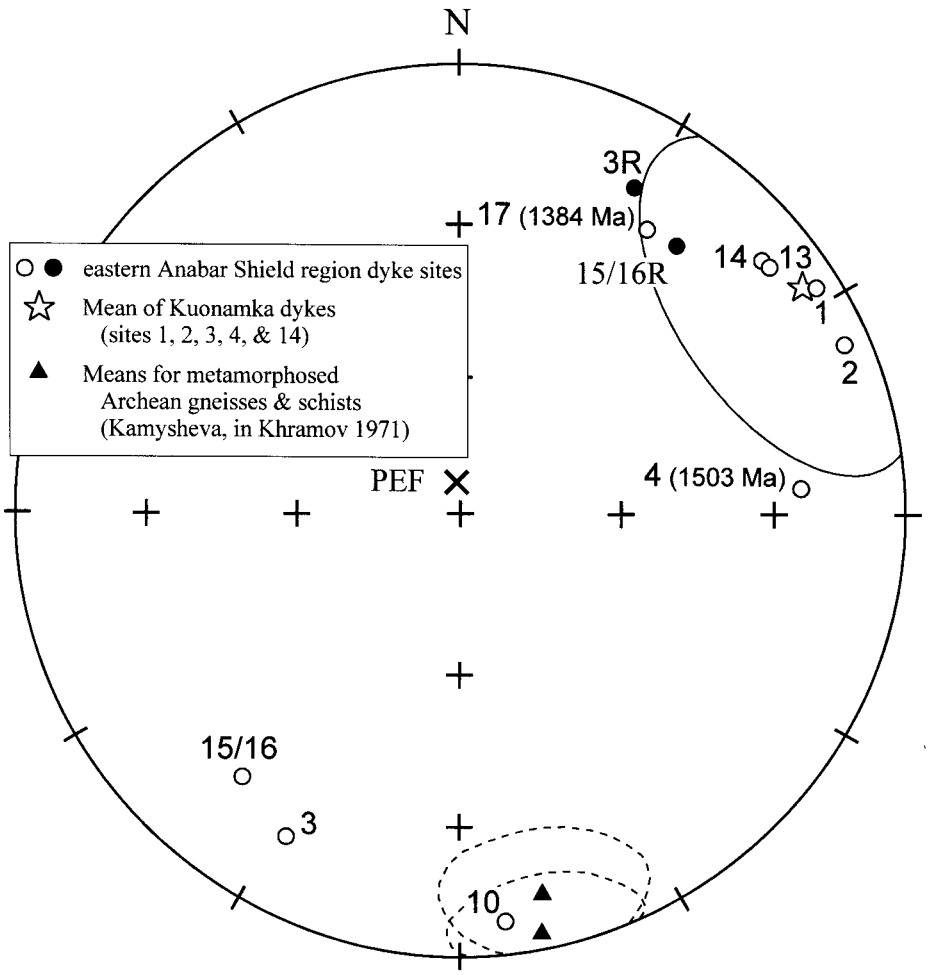


Figure 4. Paleomagnetic directions for Anabar Shield dike sites (*circles*) plotted on an equal area net with closed (open) symbols representing down (up) directions. *Star*, mean of Kuonamka sites *1*, *2*, *3R*, *4*, and *14*. The site *3R* mean is the antipode location of the site *3* mean direction. The site *1* mean is based on two reversed polarity samples and one antipodal normal polarity sample. The plotted values are from D', I' in table 1, which were recalculated for a common geographic location. Triangles are the mean directions reported by G. G. Kamysheva (in Khramov 1971) for two studies of metamorphosed Archean gneisses and schists of the Bol'shaya Kuonamka River area (see discussion in text). Ovals of 95% confidence are dashed. The "X" labeled *PEF* is the local position of present earth's field ($D = 355°$, $I = 83°$).

grouped together for statistical purposes (table 1) because they occur in the same outcrop and are thought to be parts of the feeder system for a single overlying volcanic flow and, hence, to represent a single time of emplacement.

Central Anabar Dike. Paleomagnetic results from the east-northeast-trending dike at site 10 suggest that it most likely belongs to the Central Anabar swarm (see discussion below), although its trend is also consistent with that of the Kuonamka swarm (fig. 3). Four of the six samples at this site carry a shallow southerly direction of remanence (fig. 9), quite unlike that of the Kuonamka dikes (fig. 5). Their intensities of remanent magnetization range between 1.4 and 9.1 A/m. The remaining two samples, collected a minimum of 5 m from the other four, have remanence directions that are discordant from each other and from the grouping of the other four samples and very high intensities (89 and 186 A/m), suggesting that they were struck by lightning.

East Anabar Dike. The results from the southeast-trending East Anabar dike at site 11 are exceedingly complicated, and no consistent stable

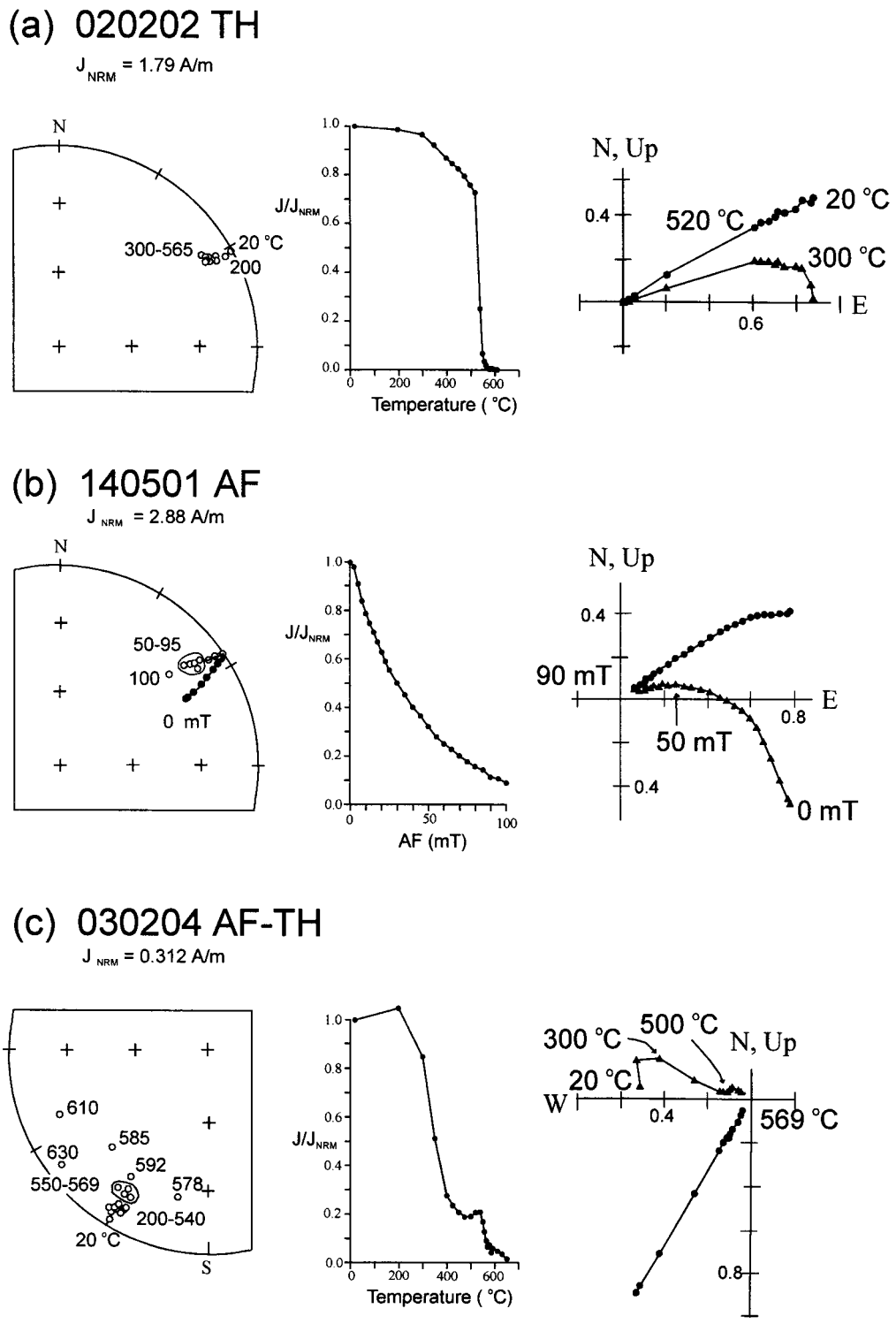


Figure 5. Examples of paleomagnetic data for Kuonamka dikes. *TH*, *AF*, and *AF-TH* indicate thermal, alternating field, and combined alternating field–thermal demagnetization, respectively. J_{NRM} is intensity of natural remanent magnetization. Closed (open) circles on the equal area net indicate down (up) direction. Circles (triangles) on the orthogonal component plot represent projection onto the horizontal (vertical) plane. In *a*, only the thermal portion of the AF-thermal experiment is shown. Samples are from (*a*) site 2, (*b*) site 14, and (*c*) site 3.

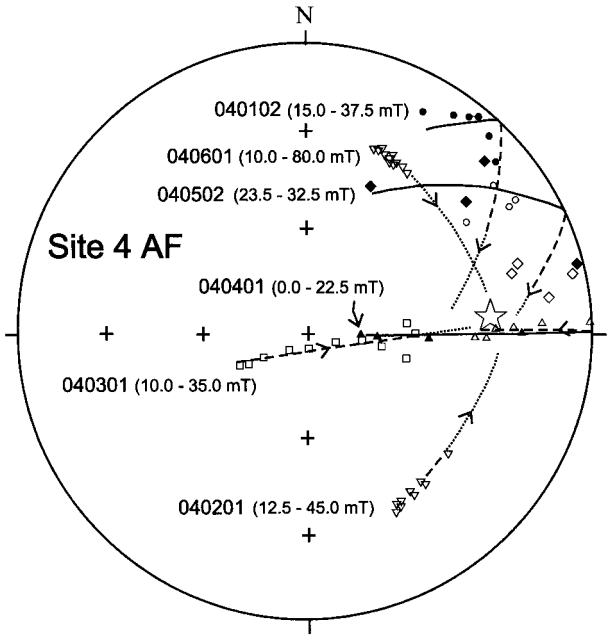


Figure 6. Paleomagnetic data for Kuonamka dike site 4 on AF demagnetization. Equal-area stereonet showing intersecting remagnetization circles for six samples. A computed great circle path through the demagnetization data is shown for each sample, with a solid (dashed) line on lower (upper) hemisphere. The path is continued beyond the data with a dotted line. Calculated intersection point of remagnetization circles is given by large open star. Closed (open) symbols on the equal area net indicate down (up) directions. For samples 040102 and 040502 the data are quite scattered along the great circle path.

endpoint was isolated. On AF demagnetization, some samples appear to reach a down southerly direction with intermediate inclinations. However, on thermal demagnetization no evidence of this component is observed. Instead, directions swing, often to shallow southerly directions with up inclinations. Stable endpoints are not attained in the thermal demagnetization experiments.

U-Pb Dating Results

Chieress Dike at Site 17. Abundant honey-brown striated blades and blade fragments of baddeleyite were recovered from the monzodioritic interior phase of the Chieress diabase at site 17, with most grains preserving lengths between 50 and 200 μm. Analytical data are reported in table 2 and are presented on a conventional concordia plot in figure 10. Four fractions of baddeleyite were analyzed from the Chieress dike, each consisting of approximately 20–30 grains. The grains were clear and unfrosted and support the observation that the baddeleyites were free of any pseudomorphing secondary zircon overgrowths. Nonetheless, all baddeleyite fractions fall in a tight cluster, between 2.6% and 3.1% discordant. As a result, they define a relatively narrow range of $^{207}Pb/^{206}Pb$ ages, between 1383.1 Ma and 1386.3 Ma, with a weighted mean age of 1384 ± 2 Ma. Because the analyses are slightly discordant and clustered, the age, which assumes zero-age Pb loss, must be considered as a minimum for crystallization of the dike. However, given the relatively low metamorphic grade of the sample (see above) and the lack of identifiable zircon overgrowths on the baddeleyites, we do not consider it likely that the true emplacement age of the Chieress diabase dike sample is significantly older than 1384 Ma.

Kuonamka Dike at Site 4. The Kuonamka dike at site 4 yielded good-quality honey-brown bladed forms of baddeleyite. Although not abundant, a sufficient number of grains were available for one precise U-Pb analysis (table 2). Data for this single fraction is ~1.7% discordant (fig. 11*a*) and yields a $^{207}Pb/^{206}Pb$ age of 1503 ± 5 Ma (again, zero-age Pb loss is assumed). Although more analyses would be helpful, datable material from this sample has been exhausted, and an improvement to the age of this dike is not currently possible. We, therefore, provisionally suggest that 1503 Ma represents the magmatic age of crystallization for this dike.

Kuonamka Dike at Site 2. From the Kuonamka dike at site 2, insufficient baddeleyite was recovered for U-Pb analysis. However, approximately 25 grains of zircon, of a variety of morphologies and clarity, were concentrated in the least-magnetic splits. The best grains in terms of igneous morphology, clarity, lack of identifiable cores or internal fractures, were selected for analysis. These comprised a heterogeneous assortment of slightly elongate to spindle-shaped, pale- to medium-brown, slightly cracked crystals with occasional synneusis twins; water-clear, colorless needles; or sharply terminated grain fragments. Following air abrasion, five fractions were chosen for analysis, comprising either single grains (B1, C2, D1) or two grains each (A1, A2). The results are presented in table 2 and figure 11*b*. Most fractions are either concordant or only slightly discordant (maximum 1.6%) at the 2σ level. They display a spread in $^{207}Pb/^{206}Pb$ ages ranging from 2679 ± 16 to 2728 ± 3 Ma. For several reasons, these ages are not interpreted to date the emplacement of the dike but rather to reflect inheritance. First, if the dike were Archean in age (ca. 2700 Ma), it would likely have been af-

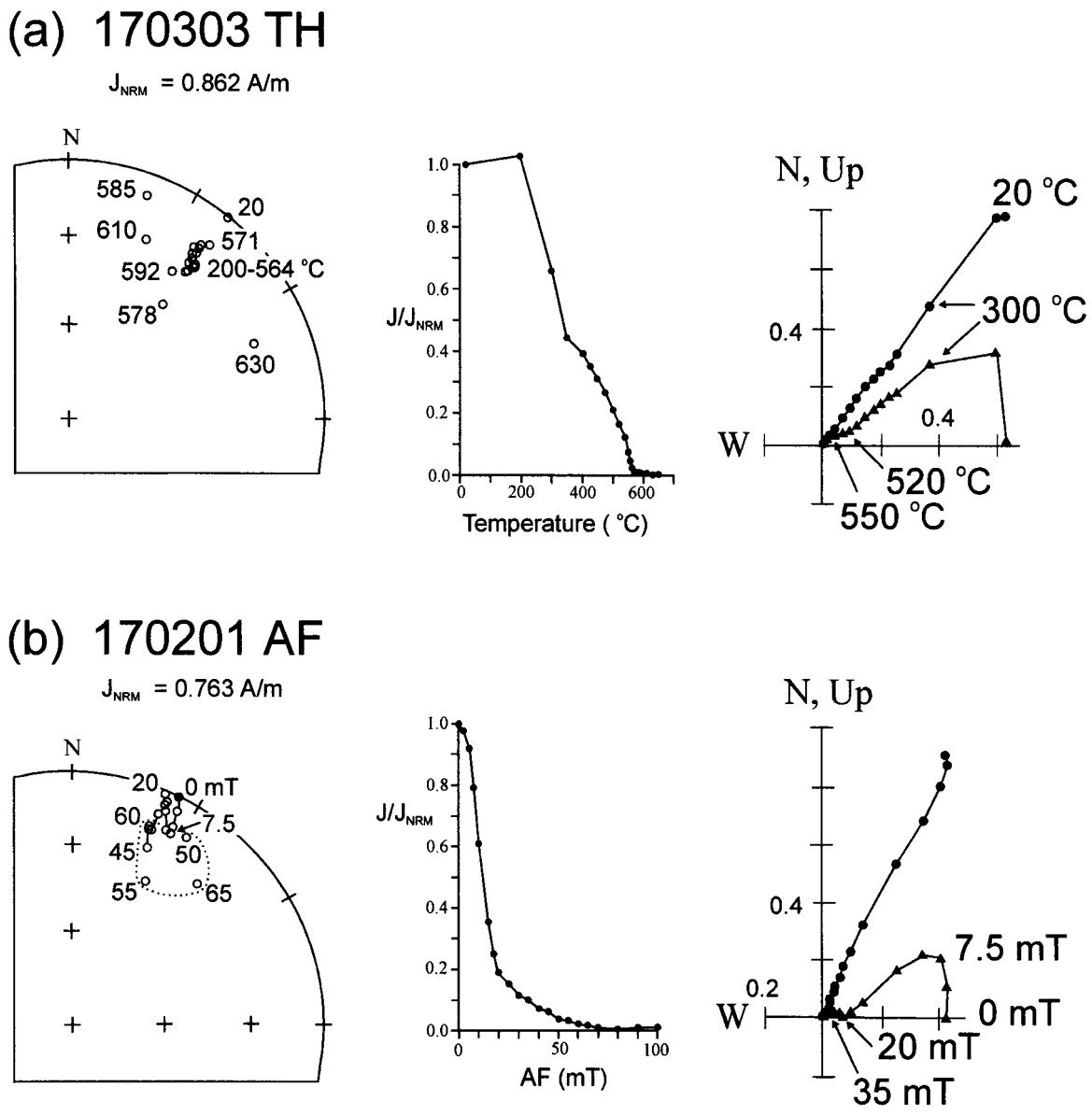


Figure 7. Examples of paleomagnetic data for Chieress dike site 17; symbols as in figure 5

fected by widespread metamorphism associated with the assembly of Siberia between 2000 and 1800 Ma (Rosen et al. 1994), and yet, petrographically the dike is unmetamorphosed. Second, the low zircon yield and the suspect inhomogeneous grain morphologies and variable quality suggest a xenocrystic rather than a magmatic origin. Third, the ~50-m.yr. spread in concordant or nearly concordant zircon data is greater than one would expect from a single magmatic zircon population from a single dike. Finally, K-Ar analysis of the dike at site 2 has yielded an age of 1242 ± 20 Ma (A. V. Okrugin, unpub. data). The ages of zircon xenocrysts within the dike could indicate that the crust underlying this part of the eastern Anabar Shield is about 2700 m.yr. old. However, anisotropy of magnetic susceptibility and geochemical studies indicate that many large dike swarms are characterized by lateral flow over much of their length (e.g., Baragar et al. 1996; Ernst and Buchan 1997).

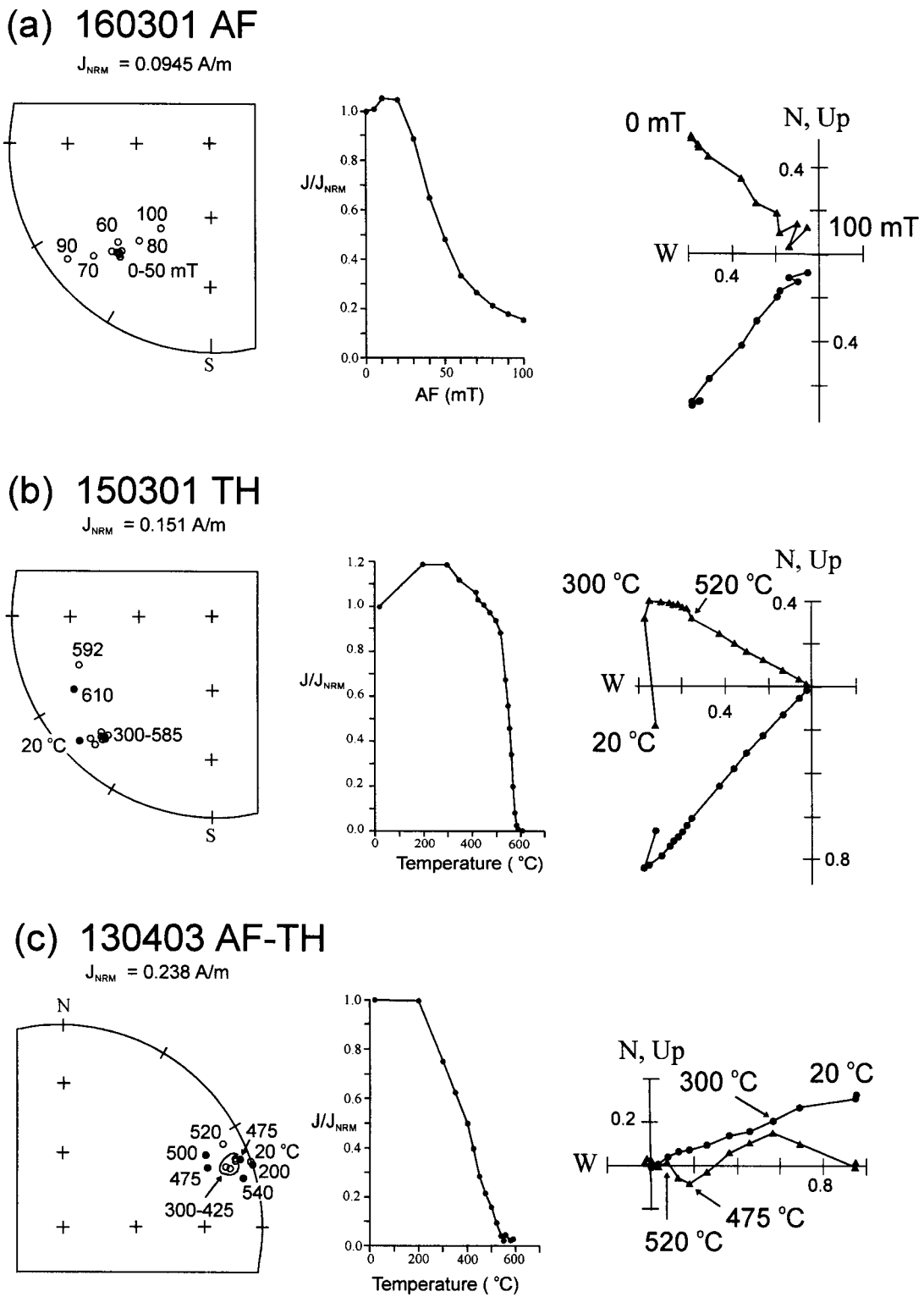


Figure 8. Examples of paleomagnetic data for Juken dikes; symbols as in figure 5. Samples are from (*a*) site 16, (*b*) site 15, and (*c*) site 13.

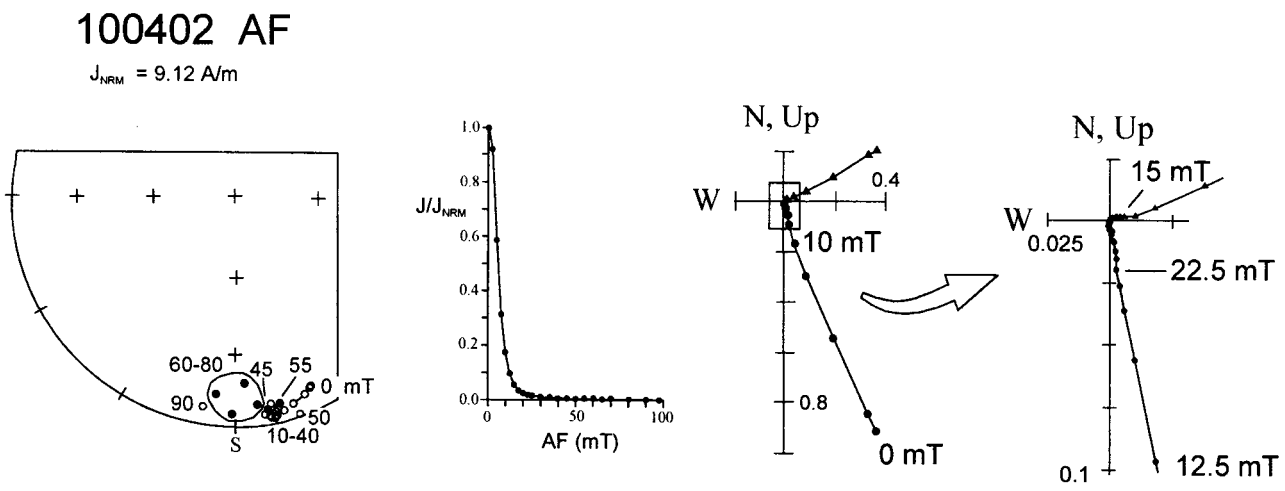


Figure 9. Examples of paleomagnetic data for possible Central Anabar dike site 10; symbols as in figure 5

Therefore, the source area(s) for the xenocrysts could be significantly east or west of the sampling area.

Age of Magnetization

Buchan and Halls (1990) discussed the importance of establishing the precise age of magnetization of diabase dikes and other mafic intrusions as a prerequisite to their use in determining reliable apparent polar wander paths and paleocontinental reconstructions. U-Pb dating of baddeleyite in this study establishes the age of emplacement of two dikes, a 1384 ± 2-Ma dike (site 17) of the Chieress swarm and, provisionally, a 1503 ± 5-Ma dike (site 4) of the Kuonamka swarm. No paleomagnetic field tests are available to rigorously establish whether the stable magnetic remanences of either of these dikes are primary or whether they were acquired as later overprints. However, there are several reasons for concluding that these dikes and, indeed, all the sampled sites assigned to the 1503-Ma Kuonamka, 1384-Ma Chieress, and ca. 1300–1200-Ma Juken swarms preserve paleomagnetic directions dating from the time of intrusion.

First, no major post-1500-Ma metamorphic event is known to have affected the Anabar Shield region (e.g., see discussion of Proterozoic orogenic events in the Siberian Plate in Rosen et al. 1994). Second, our petrographic observations of the Kuonamka, Chieress, and Juken dikes indicate low metamorphic grades. Kuonamka dikes are fresh or only slightly altered. The Chieress dike has experienced only subgreenschist metamorphism. Juken dikes are more extensively altered, but the alteration is of a low-temperature type and probably of deuteric origin. Third, the paleomagnetic directions from the Kuonamka, Chieress, and Juken dikes are different from those in the country rocks (metamorphosed Archean gneisses and schists) in the same area. Specifically, G. G. Kamysheva (in Khramov 1971) obtained remanence directions of $D = 168°$, $I = 04°$ ($\alpha_{95} = 13°$; $k = 14$) at 10 sampling sites and $D = 167°$, $I = 13°$ ($\alpha_{95} = 15°$; $k = 13$) at nine sampling sites from the vicinity of the eastern and western portions of our traverse along the Bol'shaya Kuonamka River, respectively. He interpreted these remanences as magnetic overprints acquired in the late stages of regional metamorphism, probably during the last major orogenic event in the Siberian Plate at ca. 2000–1800 Ma (Rosen et al. 1994). The paleomagnetic directions from the Kuonamka, Chieress, and Juken dikes of this study are quite different from those of the metamorphosed Archean rocks (fig. 4). This suggests that the dikes carry a primary remanence, because any major overprint that affected the dikes should also be found in the older Archean rocks.

Both normal and reversed components are observed at Kuonamka sites 1 and 3. It is unlikely that these dikes cooled over a time interval sufficiently long to encompass a field reversal. Therefore, the remanence of at least one polarity in each of these sites must have been acquired after emplacement. The mechanism of acquistion of this dual polarity remanence is unclear, but similar dual polarity results have occasionally been reported in other diabase dike swarms whose remanence is known, on the basis of paleomagnetic field tests, to be primary. For example, Matachewan dikes of

Table 2. U-Pb Results for Chieress and Kuonamka Diabase Dikes, Eastern Anabar Shield Region

Fraction	Weight (μg)	U (ppm)	Pb* (ppm)	$^{206}Pb/^{204}Pb$[a]	Pb (pg)[b]	$^{208}Pb/^{206}Pb$[a]	Radiogenic ratios ($\pm 1\sigma$, %) $^{206}Pb/^{238}U$[c]	$^{207}Pb/^{235}U$[c]	$^{207}Pb/^{206}Pb$[c]	$^{207}Pb/^{206}Pb$ age (Ma)[d]	Disc. (%)[e]
Sample EQ94-17 (Chieress), baddeleyite:											
A1, NM5	21	78	17	5491	4	.0152	.2332 ± .10	2.835 ± .11	.08818 ± .04	1386.3 ± 1.5	2.8
A2, NM5	13	221	49	2994	14	.0288	.2329 ± .10	2.831 ± .13	.08818 ± .07	1386.3 ± 2.5	2.9
B1, M5	49	246	55	13,770	13	.0312	.2330 ± .09	2.828 ± .10	.08803 ± .03	1383.1 ± 1.1	2.6
B2, M5	49	158	35	7062	16	.0307	.2321 ± .09	2.819 ± .10	.08807 ± .03	1384.0 ± 1.3	3.1
Sample EQ94-4 (Kuonamka), baddeleyite:											
A, NM5	…	…	…	1533	3	.0393	.2581 ± .14	3.336 ± .14	.09375 ± .13	1503.0 ± 4.8	1.7
Sample EQ94-2 (Kuonamka), zircon:											
A1, NM0	2	45	26	143	23	.1006	.5149 ± .48	12.983 ± .72	.18287 ± .48	2679.0 ± 16.0	.1
A2, NM0	2	105	62	409	17	.1551	.5139 ± .21	13.183 ± .27	.18605 ± .16	2707.6 ± 5.1	1.6
B1, NM0	1.5	20	11	135	8	.0877	.5132 ± 1.35	13.052 ± 1.39	.18444 ± .45	2693.2 ± 15.0	1.0
C2, NM2	.5	150	97	946	3	.2485	.5223 ± .19	13.564 ± .18	.18833 ± .09	2727.7 ± 2.9	.8
D1, NM2	4	43	28	1927	3	.2639	.5287 ± .14	13.723 ± .13	.18824 ± .07	2726.8 ± 2.4	−.4

Note. "NM" and "M" refer to nonmagnetic and magnetic splits, respectively, at maximum current and 5°, 2°, or 0° side slope on a Frantz isodynamic separator. Pb* = radiogenic Pb. Analytical techniques employing a mixed $^{205}Pb/^{233-235}U$ spike detailed in Parrish et al. (1987). Total procedural blanks were 2–6 pg for Pb and 1–2 pg for U. Error propagation follows that described by Roddick (1987).

[a] Measured ratio, corrected for mass fractionation and spike.

[b] Total common Pb in analysis in picograms, corrected for fractionation and spike.

[c] Corrected for spike, blank, and common Pb.

[d] Errors for age are ±2 S.E. in Ma.

[e] Discordance along a chord to the origin.

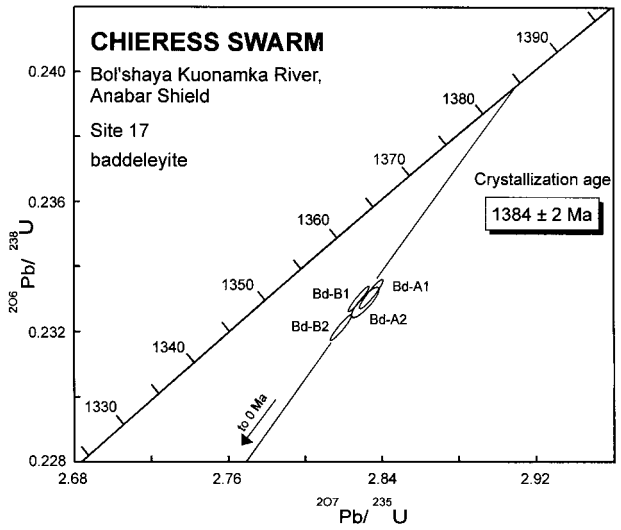


Figure 10. Concordia diagram showing U-Pb baddeleyite age for Chieress site 17.

the southern Canadian Shield sometimes carry both polarities at a single sampling site (e.g., Ernst and Halls 1984, p. 1503 and fig. 3), perhaps as the result of a self-reversal or chemical alteration shortly after emplacement.

The dike at site 10 is interpreted to carry a magnetic overprint and to belong to the Central Anabar swarm. It has a paleomagnetic direction similar to that obtained from the host Archean gneisses and schists (fig. 4). This suggests that site 10 could have been magnetized at the same time as the Archean rocks, perhaps during the 2000–1800-Ma metamorphic event (Rosen et al. 1994). However, since the dike is undeformed and the mineralogy of site 10 is fresh, the dike must have been emplaced and magnetized during the late stages of this event. If so, site 10 may belong to the Central Anabar swarm (fig. 2) that has K-Ar ages as old as ca. 1850 Ma (fig. 1 of Okrugin et al. 1990).

Discussion

The paleolatitude and azimuthal orientation of the Siberian Plate at 1503 and 1384 Ma, based on the paleopoles determined in this study from the eastern Anabar Shield region (table 1), are illustrated in figure 12*a*. For comparison, the early Mesoproterozoic positions based on paleomagnetism for Laurentia are shown in figure 12*b*. It is assumed that the Siberian Plate has remained intact since the Mesoproterozoic, except for the mid-Paleozoic opening of the Viljuy Rift, although this assumption has yet to be rigorously tested with paleomagnetic data from different parts of the Siberian Plate.

In the drift plots of figure 12, paleolongitude is not constrained by the paleomagnetic data. In addition, because of ambiguity in magnetic polarity, a craton can generally be placed in either hemisphere with its orientation differing by 180° (e.g., compare the two positions of Siberia at 1384 Ma in fig. 12*a*). However, there is no such polarity ambiguity in the relative positions of a craton at different times if the time intervals between paleomagnetic data are too short (<100 m.yr.) for the craton to have rotated through 180° or to have drifted over the geographic pole to the opposite side of the globe.

The virtual geomagnetic pole (VGP) for the 1503-Ma Kuonmaka dike at site 4 falls at 16°S, 221°E (dm = 17°, dp = 10°), corresponding to a paleolati-

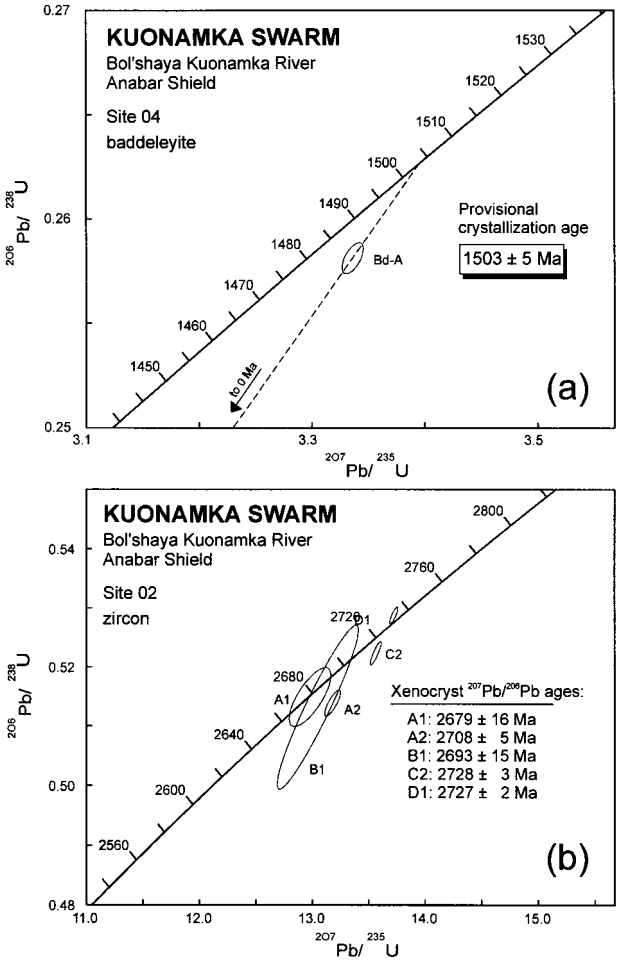


Figure 11. Concordia diagrams for Kuonamka swarm U-Pb geochronology. *a*, Baddeleyite crystallization age for site 2. *b*, Zircon xenocryst ages for site 4.

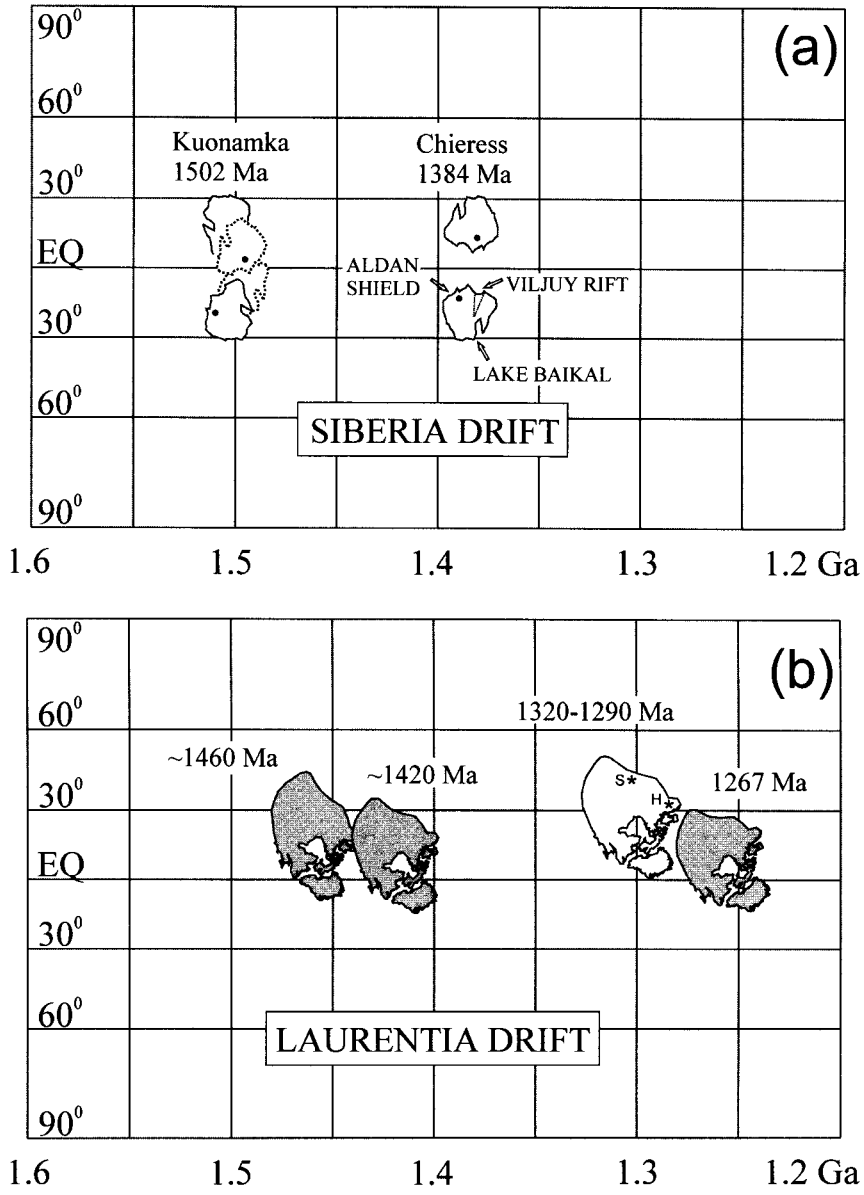


Figure 12. Comparison of latitudinal drift and rotation of Siberia and Laurentia in the early Mesoproterozoic based on paleomagnetism. *a*, Solid outlines of Siberia at 1503 and 1384 Ma are based on virtual geomagnetic poles (VGPs) from sites 4 and 17 of this study. Dashed outline of Siberia is based on the paleopole calculated from all probable Kuonamka sites, 1, 2, 3, 4, and 14. Viljuy Rift of Siberia should be closed between the dotted lines to take into account the ~20° opening thought to have occurred in the mid-Paleozoic (see text). The location of Anabar Shield is given by a filled circle, and the location of Lake Baikal is labeled (see fig. 2 for detailed outline of Siberia and full labeling of geographic features). *b*, Greenland has been restored to its late Mesozoic position relative to the rest of Laurentia based on seafloor magnetic anomalies in the Labrador Sea (Roest and Srivastava 1989). Positions of Laurentia are based on published paleomagnetic poles that are discussed in the text. All of these poles are from rock units that have been dated by the U-Pb method. Those in gray are derived from "key" paleopoles that have been demonstrated primary with a paleomagnetic field test (Buchan et al. 2000). Location of 1.38-Ga Hart River sills of northwest Laurentia and the 1.38-Ga Salmon River sills of the Belt-Purcell Basin of western Laurentia are shown as stars labeled *H* and *S*, respectively. Because of the polarity ambiguity in paleomagnetic data, a given craton can be placed in either hemisphere as described in the text. Therefore, with Laurentia in one hemisphere as shown in *b*, Siberia can be located in either hemisphere for the purposes of comparison. Longitude is unconstrained by the paleomagnetic data. Cratons were positioned using GMAP for Windows (version 3.0).

tude of 19° ± 10°. However, this can only be considered a rough approximation because data from a single dike do not average out magnetic secular variation. A somewhat better estimate of the paleolatitude and azimuthal orientation of the Siberian Plate at 1503 Ma may be obtained from the mean paleopole for all five sites (1, 2, 3, 4, and 14) that are assigned to the Kuonamka swarm. This Kuonamka mean paleopole is 6°N, 234°E (dm = 28°, dp = 14°), and the corresponding paleolatitude is 5° ± 14°. In this case, much of the magnetic secular variation is likely to have been averaged out.

The paleomagnetic direction for the 1384-Ma Chieress dike at site 17 corresponds to a VGP at 4°N, 258°E (dm = 9°, dp = 5°) and a paleolatitude for the sampling locality of 12° ± 5°. Secular variation is not averaged out.

At both 1503 and 1384 Ma the Siberian Plate is in low latitudes (fig. 12*a*). Between 1503 (using the position based on the mean Kuonamka paleopole) and 1384 Ma, the craton appears to have rotated ~25° clockwise. However, it should be noted that this rotation is not statistically significant given the uncertainties associated with the paleo-orientation of the poles total at least 38° (see note to table 1).

No ca. 1503- or 1384-Ma paleopoles are currently available for Laurentia, but there are well-dated paleopoles for ca. 1460–1420, 1320–1290, and 1267 Ma (fig. 12*b*; see paleomagnetic and U-Pb age tabulation in Buchan et al. 2000). Paleopoles for the 1460–1420-Ma plutons of northeastern and southwestern Laurentia are thought to be primary on the basis of a positive baked contact test for the ca. 1460-Ma Michikamau anorthosite. The other paleopoles of this group are from the ca. 1450-Ma Harp Lake Complex, the ca. 1434-Ma Laramie anorthosite and Sherman granite, and the ca. 1420-Ma Mistastin pluton. The 1320–1290-Ma paleopole is from the Nain anorthosite of northeastern Laurentia. These units are all large plutons that cooled slowly, so that their primary magnetic remanences may have been acquired up to several million years after the U-Pb emplacement ages (e.g., Harlan et al. 1994). The 1267-Ma pole is from the Mackenzie dike swarm of northern Laurentia.

All the above paleopoles locate Laurentia at low latitudes with a fairly consistent azimuthal orientation (fig. 12*b*). As noted above, the preliminary paleomagnetic data for Siberia in this study also places Siberia at low latitudes at ca. 1503 and 1384 Ma. This is consistent with Siberia and Laurentia moving together in the early Mesoproterozoic. Thus, a position for Siberia adjacent to the northern margin of Laurentia (see fig. 1) is permitted by the paleomagnetic data. However, other locations for Siberia adjacent to Laurentia are equally permissible given that the ages of data from the two cratons are not well matched, there are rather large uncertainties in the paleolatitudes and paleo-orientations of Siberia because of the scatter in the Kuonamka directions, and the Chieress and Kuonamka results do not average out secular variation. In addition, because of the paleolongitude uncertainty inherent in the paleomagnetic method, the data would allow the two continents to have been apart, provided they were both in relatively low latitudes through the period in question. Additional well-dated paleomagnetic data are required if the position of Siberia is to be accurately determined in the Mesoproterozoic.

The relationship between Siberia and Laurentia in the Neoproterozoic is also unclear, despite a recent paleomagnetic model of Smethurst et al. (1998) that locates Siberia either adjacent to Baltica (fig. 1*h*) or adjacent to East Antarctica in a ca. 750-Ma Rodinia reconstruction. This fit is based on a comparison of well-dated paleopoles for Laurentia (summary in Torsvik et al. 1996) at ca. 750 Ma and the paleopole for gabbro sills that intrude the Sette-Daban fold belt of southeastern Siberia (Pavlov et al. 1992; Pavlov 1996). Smethurst et al. (1998) assigned an age of 730 Ma to these sills based on K-Ar dating and geological constraints. However, they are now known to be much older than previously thought. Pavlov et al. (1992) gave an Sm-Nd age of 942 ± 18 Ma for the sills in their paleomagnetic study. Rainbird et al. (1998) reported minimum emplacement ages of 974 ± 7 Ma and 1005 ± 4 Ma based on U-Pb baddeleyite dating for two sills located about 150 km to the north. These ages are about 250 m.yr. older than the age assigned to the sills by Smethurst et al. (1998) and invalidate their paleomagnetic reconstruction (fig. 1*h*).

Unfortunately, there are several problems that prevent the use of the paleopole from the Sette-Daban sills for reliable reconstructions of Siberia and Laurentia at ca. 1000 Ma. First, the very small degree of scatter (α_{95} = 2.7° and k = 426 for eight sill sites) suggests that secular variation has not been averaged out. Second, although a fold test suggests that the remanence in the sills predates Mesozoic folding (Pavlov et al. 1992), siltstones and sandstones of the host Kandyk suite yielded a magnetization direction similar to that obtained from the sills (Pavlov et al. 1992). This suggests that the sediments and sills differ little in age or that they carry a magnetic overprint. Third, a direct comparison with Laurentia paleopoles is difficult because of rapid continental drift of Laurentia at ca. 1050–950

Ma (see Grenville track of Buchan et al. 2000; fig. 2*c*) and a lack of precise dating of the Laurentia paleopoles of this age.

If large flood basalt provinces have been fragmented by continental drift, then precise dating of the components on different continents may help to reconstruct their configuration at the time of emplacement. For example, Rainbird et al. (1998) implied that a strong test of reconstructions that place Siberia adjacent to northern Laurentia would be the identification in Siberia of the extensive 1267-Ma Mackenzie and 723-Ma Franklin mafic magmatic events of northern Laurentia (Ernst and Buchan 1997 and references therein). As yet units of these ages have not been found in Siberia.

However, the 1384-Ma Chieress dike age from Siberia is almost identical with that of two diabase sill units in northwestern and western Laurentia (locations in fig. 12*b*). Hart River sills intrude the Paleoproterozoic Wernecke Supergroup in the Ogilvie Mountains of the northern Cordillera and have U-Pb zircon ages of 1380 ± 4, 1385 ± 2, and 1383 ± 6 Ma (Abbott 1997). Also, a sill belonging to the large Salmon River bimodal mafic complex at the base of the Belt-Purcell Supergroup of the Belt Basin has been dated at 1379 ± 1 Ma (U-Pb zircon; Doughty and Chamberlain 1996). The implications of the Chieress–Hart River–Salmon River age correlation are unclear. It could be argued that this correlation supports models of Rodinia in which Siberia is located adjacent to northwestern or western Laurentia. However, Baltica also contains mafic intrusive units with U-Pb ages of 1380–1360 Ma (Åhäll and Connelly 1998) and has been reliably positioned adjacent to northeastern Laurentia at 1265 Ma on the basis of identical aged paleopoles from the two cratons and geological considerations (summary in Buchan et al. 2000). Diabases in northwestern Tanzania and southeastern Burundi have also been dated at 1380–1350 Ma using the ^{40}Ar-^{39}Ar technique (Deblond et al. 1999). Such a widespread distribution of ca. 1380 Ma events on four continents may require more than a single magmatic event at that time.

Conclusions

Diabase dikes that were sampled in the eastern Anabar Shield region are interpreted to belong to four Mesoproterozoic and two Paleoproterozoic swarms on the basis of paleomagnetism, U-Pb or previous K-Ar geochronology, trend, and petrography. Five east-trending dikes are assigned to the Kuonamka swarm, dated herein by the U-Pb method at 1503 ± 5 Ma. A single dike from the Chieress swarm yielded a U-Pb age of 1384 ± 2 Ma. Of the remaining five dikes, three are thought to represent the ca. 1300–1200-Ma Juken swarm, and the fourth and fifth are assigned to the ca. 1800–1400-Ma Central Anabar and ca. 1700-Ma Eastern Anabar swarms.

Paleomagnetic data from the 1503-Ma Kuonamka and 1384-Ma Chieress swarms locate the Anabar Shield, and perhaps the whole Siberian Plate, at low latitudes during the early Mesoproterozoic. Well-constrained paleomagnetic data for Laurentia also place that craton at low latitudes at 1460–1420, 1320–1290, and 1267 Ma. Thus, these data would allow Laurentia and Siberia to have drifted together in the Mesoproterozoic, as proposed by several recent studies using geological criteria. However, the ages of the paleomagnetic data from the Anabar Shield and Laurentia are not well matched, the paleomagnetic data from the Anabar Shield do not average out secular variation, and it remains to be confirmed that the Anabar Shield paleomagnetic data apply to the Siberian Shield as a whole. Therefore, it is not yet possible to propose a reliable early Mesoproterozoic reconstruction or to discriminate between the different reconstruction options for Siberia and Laurentia summarized in figure 1.

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